

Econometric_HW0505

#11.24(a)

Reduced-Form Equation for lprice	
Dependent variable:	
	lprice
mon	-0.108 (0.103)
tue	-0.066 (0.101)
wed	-0.049 (0.104)
thu	0.039 (0.101)
stormy	0.446*** (0.079)
mixed	0.237*** (0.079)
Constant	-0.360*** (0.079)
Observations	111
R2	0.245
Adjusted R2	0.201
Residual Std. Error	0.341 (df = 104)
F Statistic	5.624*** (df = 6; 104)
Note: *p<0.1; **p<0.05; ***p<0.01	

Significance of Instruments for Demand		
	Variable_Test	Value
1	MIXED (p-value)	0.003
2	Joint F-stat (STORMY, MIXED)	16.140

The variable MIXED has a p-value of **0.003** < 0.05, it is statistically significant at the 5% level. Testing the joint significance of STORMY and MIXED yields an F-value of **16.14** > 10, these variables serve as **strong instruments** for estimating the demand equation.

#11.24(b)

IV Estimation of Demand Equation	
Dependent variable:	
lquan	
lprice	-0.930*** (0.353)
mon	-0.012 (0.208)
tue	-0.526** (0.202)
wed	-0.563*** (0.207)
thu	0.100 (0.203)
Constant	8.540*** (0.157)
Observations	111
R2	0.185
Adjusted R2	0.146
Residual Std. Error	0.685 (df = 105)
Note: *p<0.1; **p<0.05; ***p<0.01	

TABLE 11.5 2SLS Estimates for Fish Demand				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
<i>C</i>	8.5059	0.1662	51.1890	0.0000
<i>ln(PRICE)</i>	-1.1194	0.4286	-2.6115	0.0103
<i>MON</i>	-0.0254	0.2148	-0.1183	0.9061
<i>TUE</i>	-0.5308	0.2080	-2.5518	0.0122
<i>WED</i>	-0.5664	0.2128	-2.6620	0.0090
<i>THU</i>	0.1093	0.2088	0.5233	0.6018

Comparison with Table 11.5: By adding MIXED as an additional instrumental variable in part (b), the estimated coefficient for $\ln(\text{PRICE})$ changes from -1.1194 to **-0.930**. Furthermore, using more instruments typically improves estimation efficiency. We can observe this as the standard error for $\ln(\text{PRICE})$ decreases from 0.4286 (in Table 11.5) to **0.353**.

#11.24(c)

Surplus Instrument Validity	
Test	p_value
1 Sargan (Overidentification)	0.380

The Sargan test yields a p-value of **0.38** > 0.05, we fail to reject the null hypothesis and conclude that the surplus instrument is **valid** (meaning the instruments are effectively uncorrelated with the structural error term of the demand equation).

#11.24(d)

Significance of Instruments for Supply		
Test	F_statistic	p_value
1 Joint F-test (MON to THU)	0.628	0.644

In the reduced-form equation for price, the joint F-statistic for MON, TUE, WED, and THU is **0.628** with a p-value of **0.644**. These day-of-the-week indicator variables are the necessary instrumental variables to identify the **Supply** equation. The F-statistic is significantly less than 10, it indicates these instruments are **weak**. Consequently, we **cannot** estimate the supply equation by 2SLS with confidence, as weak instruments would lead to biased and unreliable estimates.

#11.28(a)

Demand

$$Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$$

$$\alpha_2 P_i = Q_i - \alpha_1 - \alpha_3 PS_i - \alpha_4 DI_i - e_{di}$$

$$P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di}$$

預期符號：

- $1/\alpha_2 < 0$ ，因為需求曲線為負斜率， $\alpha_2 < 0$
- $-\alpha_3/\alpha_2 > 0$ ，因為替代品價格與需求數量為正相關， $\alpha_3 > 0$
- $-\alpha_4/\alpha_2 > 0$ ，因為可支配所得越高，需求數量越高， $\alpha_4 > 0$

Supply

$$Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$$

$$\beta_2 P_i = Q_i - \beta_1 - \beta_3 PF_i - e_{si}$$

$$P_i = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si}$$

- $1/\beta_2 > 0$ ：因為供給曲線為正斜率， $\beta_2 > 0$
- $-\beta_3/\beta_2 > 0$ ：因為要素成本越高，售價也越高， $\beta_3 < 0$

#11.28(b)

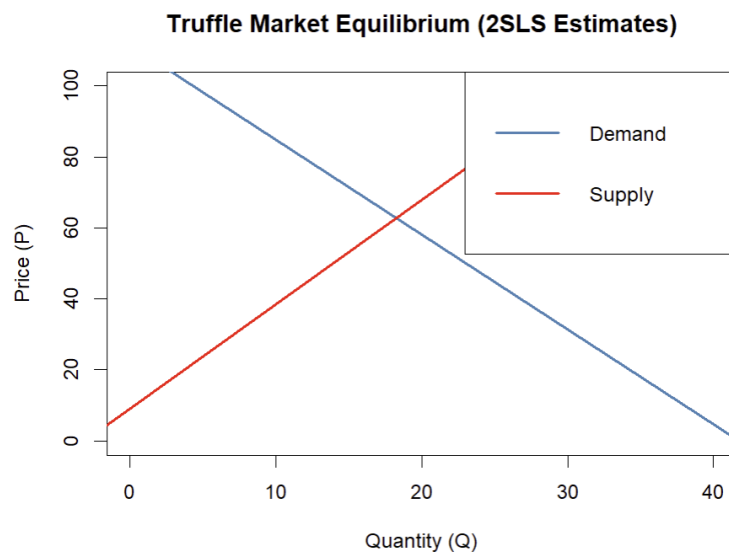
2SLS Estimates for Inverse Demand and Supply		
Dependent variable:		
	p	
	Inverse Demand (1)	Inverse Supply (2)
q	-2.671** (1.175)	2.937*** (0.216)
ps	3.461*** (1.116)	
di	13.390*** (2.747)	
pf		2.958*** (0.156)
Constant	-11.428 (13.592)	-58.798*** (5.859)
Observations	30	30
R2	0.557	0.949
Adjusted R2	0.506	0.945
Residual Std. Error	13.166 (df = 26)	4.399 (df = 27)
Note: *p<0.1; **p<0.05; ***p<0.01		

Based on the IV output, the sign of Q in the demand equation is **negative** and in the supply equation is **positive**. These results match expectations from (a). The coefficients for Q are significantly different from zero at the 5% level.

#11.28(c)

Item	Value
q Price Elasticity of Demand (at means)	-1.272

#11.28(d)



#11.28(e)

Equilibrium Comparison			
Method		Equilibrium_Q	Equilibrium_P
(Intercept)	2SLS Structural	18.250	62.843
1	Reduced-Form	18.260	62.815

Conclusion: They agree perfectly. This is a mathematical property of linear simultaneous equations systems: predicting from the reduced-form yields the exact same equilibrium as solving the 2SLS structural equations.

#11.28(f)

Dependent variable:				
	p			
	OLS	instrumental	OLS	instrumental
	OLS Dem	variable	OLS Sup	variable
	(1)	IV Dem	(3)	IV Sup
	(1)	(2)	(3)	(4)
q	0.151 (0.499)	-2.671** (1.175)	2.661*** (0.171)	2.937*** (0.216)
ps	1.361** (0.594)	3.461*** (1.116)		
di	12.358*** (1.825)	13.390*** (2.747)		
pf			2.922*** (0.148)	2.958*** (0.156)
Constant	-13.619 (9.087)	-11.428 (13.592)	-52.876*** (5.024)	-58.798*** (5.859)
Observations	30	30	30	30
R2	0.801	0.557	0.953	0.949
Adjusted R2	0.778	0.506	0.950	0.945
Residual Std. Error	8.814 (df = 26)	13.166 (df = 26)	4.202 (df = 27)	4.399 (df = 27)
F Statistic	34.952*** (df = 3; 26)		274.351*** (df = 2; 27)	
Note: *p<0.1; **p<0.05; ***p<0.01				

Demand Equation: The sign for Q is **incorrect**. OLS yields a positive coefficient (0.151), which contradicts the expected negative sign derived in part (a) (the law of demand). Furthermore, this coefficient is not significantly different from zero.

Supply Equation: The sign for Q is **correct**. OLS yields a positive coefficient (2.661), which matches the expected positive sign derived in part (a) and is statistically significant.

Using OLS leads to severe **simultaneous equations bias** due to the endogeneity between price and quantity. This inconsistency is clearly seen in the OLS demand equation, which incorrectly produces a positive, insignificant slope for quantity. The 2SLS (IV) method corrects this bias, yielding the theoretically expected negative demand slope and demonstrating that OLS is inappropriate for estimating structural supply and demand equations.